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ASSIGNMENT 1

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REQUIREMENTS ANALYSIS REPORT: STEMMate Namibia

1. Problem Frame and Scope

1.1. Problem Statement and Context

The STEMMate Namibia project addresses the critical lack of structured support for adult STEM facilitators operating in resource-constrained Namibian environments. These facilitators typically work in settings characterized by intermittent connectivity, high data costs, and a reliance on shared or low-specification hardware E5. Furthermore, facilitators must adapt activities based on the immediate availability of inexpensive or local materials E1. The core challenge is to provide a reliable, offline-ready tool that enables consistent session planning without requiring constant cloud access. Success will be measured by:

- **Synchronization Reliability:** Data synchronization once a connection is established should have a 90% success rate
- **Efficiency:** The planning process should be 30% faster compared to the existing method (manual / messaging
- **Compliance:** 100% adherence to data minimization principles concerning pupils' information

1.2. System Boundary

STEMMate Namibia is a system for facilitating offline session planning by adult facilitators. Scope of the system is limited to finding STEM activities, managing session preparation and data synchronization. System boundary covers local data storage and synchronization gateway. Excluded components include pupil interfaces, school administration management, and real-time communication tools.

1.3. User and Beneficiary Definition

- **Primary Users:** STEM Facilitators and Outreach Coordinators who use the system to plan activities and maintain records.
- **Supporting Users:** Volunteer Assistants and Equipment Custodians who use the system to review safety procedures E3 or track kit status E4.
- **Secondary Stakeholders:** School Managers that require confirmation of appropriate activities E6.
- **Beneficiaries:** Pupils that participate in STEM activities. Notably, pupils do not use the system. The system facilitates Outreach Coordinator's ability to gather total numbers of participants grouped by certain categories (gender, age group, etc.) E2 but does not collect any pupil identifiable data.

1.4. Constraints and Exclusions

Operating Constraints	Explicit Exclusions
Intermittent connectivity and high data costs E5.	No pupil accounts or personal identifiers (PII) E2.
Shared or low-specification hardware E5.	No Artificial Intelligence (AI) for learner assessment.
Requirement for locally available materials E1.	No national production platform or full LMS.
High-contrast UI and keyboard support needs.	| No real-time video or high-bandwidth streaming.

2. Stakeholder Analysis

2.1. Stakeholder Mapping

Stakeholder	Role	Interest	Influence
STEM Facilitator	Primary User	High	High
Outreach Coordinator	Administrative User	High	Medium
Volunteer Assistant	Supporting User	Medium	Low
Equipment Custodian	Supporting User	Medium	Medium
School Management	Secondary Stakeholder	Medium	Medium
Pupils	Beneficiary	High (Impact)	Low (non-user)

2.2. Stakeholder Tension

A significant tension exists between the Equipment Custodian's need for rigid, real-time kit tracking and the Facilitator's operational reality of offline planning. The Custodian requires visibility into kit availability and returns status to manage shared resources across groups E4. However, the Facilitator often prepares sessions in areas without cellular signals E1, E5. If the system mandates a live connection to "check out" a kit, it creates a bottleneck that prevents the Facilitator from completing a plan. Conversely, an offline-only approach leads to scheduling conflicts for the Custodian. The architectural trade-off requires an asynchronous "request queue" where kit status is reserved locally and resolved via conflict logic during synchronization, prioritizing the Facilitator's ability to plan while accepting some temporal delay in the Custodian's oversight.

3. Ethical Validation Plan

3.1. Research Design

Validation of offline-to-online transition and shared device security will be done. The test subjects will be strictly limited to adult Facilitators and Coordinators. No contact with schools, children and their parents will take place, observing non-negotiable ethical boundary of the project.

3.2. Ethical Methodology

Interviews and usability testing will be done with adults only.

- **Informed Consent:** Participants should sign informed consent forms with a description of their ability to withdraw
- **Data Minimization:** There would be no access to school records or children's data. Only aggregate participation counts (non-identifiable) will be used in order to test the hypothesis E2.
- **Secure Data Handling:** All data provided by participants will be anonymized.
- **Ethical Safeguard:** The "Systems Architect" role ensures the system prevents the entry of PII through hard-coded field restrictions, even during testing

3.3. Validation Questions

1. Does the system effectively prevent session finalization if safety notes E3 or inclusion prompts are left blank?
2. How easy is for a facilitator to resolve a synchronization conflict when a kit was reserved by another user when he/she was offline?

3. Does the "Secure Sign-Out" and automated timeout functionality protect coordinator data on shared devices E5 without disrupting the flow of a STEM session?

4. User Understanding

4.1. Proto-Personas

Persona 1: The Facilitator

- **Archetype:** The Resourceful Educator.
- **Context:** Worked in rural centres using a shared, low-RAM smartphone E5.

- **Digital Confidence:** moderate to high; depends on digital tools but is sensitive to data charges
- **Needs:** The need to filter activities according to the availability of materials in the room E1 and stable offline availability of saved plans E5. ASSUMPTION
- **Assumption:** Prefers an icon-heavy interface to minimize language barriers ASSUMPTION.

Persona 2: The Volunteer

- **Archetype:** The Safety-Conscious Assistant.
- **Context:** Assists in session setup; often has very limited time before learners arrive E3. ● **Digital Confidence:** Moderate; requires clear, step-by-step guidance.
- **Needs:** Explicit safety warnings and preparation checklists that are visible at a glance E3.
- **Assumption:** Often views the app while the device is in a fixed position on a desk ASSUMPTION

.**4.2. Usage Scenarios**

Scenario 1: Offline Material Pivot
A facilitator arrives at a remote community center with no signal. They open STEMMate Namibia to prepare for a session. They find that the batteries expected for the "Circuit Fun" activity are missing. Using the offline-cached material filter E1, they quickly switch to a "Cardboard Structures" activity that only requires local materials. They save this new session plan locally E5, including inclusion prompts for learners with limited mobility ASSUMPTION.

Scenario 2: Shared Device Security Handover
Before a session begins, a facilitator reviews safety steps with a volunteer E3 on a shared phone. After the briefing, the facilitator uses the "Secure Sign-Out" E5 so that the phone can be safely left on a supply table. Later, the School Manager asks to see the plan E6. The facilitator signs back in, uses the "Export/View" function to show the age-appropriate steps E6, and then the system's automated timeout triggers after five minutes of inactivity to protect the account ASSUMPTION.

4.3. Evidence vs. Assumptions
Facts derived from E1–E6 are noted with brackets (e.g., E1). Inferences regarding specific UI behaviors, like icon-heavy interfaces or automated timeouts, are labelled as ASSUMPTION.

5. Prioritized Requirements

5.1. Functional Requirements (FR)

- **FR-01:** The system shall allow users to browse and filter activities by level, duration, topic, and material availability E1.
- **FR-02:** The system shall provide a "Save for Offline" function that caches all activity text and images locally.

- **FR-03:** The system shall allow facilitators to create session plans including steps, timing, and mandatory safety notes E3.
- **FR-04:** The system shall provide inclusion prompts during the session planning phase to ensure activities are adaptable E6.
- **FR-05:** The system shall allow the recording of aggregate pupil counts by broad participation category (no PII) E2.
- **FR-06:** The system shall allow users to record the return status of shared equipment kits E4.
- **FR-07:** The system shall synchronize all locally saved plans and kit data when internet connectivity is detected E5.
- **FR-08:** The system shall provide a "Manager View" or exportable summary for school oversight E6.

5.2. Quality Requirements (QR)

- **QR-01:** The system must load and be interactive within 5 seconds on devices with 2GB RAM or less.
- **QR-02:** The system must utilize a local-first database to ensure zero data loss during signal drops E5.
- **QR-03:** The UI must maintain a minimum contrast ratio of 4.5:1 and support keyboard only navigation.

5.3. Specialized Requirements

- **Privacy (PR-01):** The system shall strictly prohibit the entry of pupil names, photos, or school IDs; data entry is limited to aggregate counts E2.
- **Security (SEC-01):** The system shall feature a prominent "Secure Sign-Out" button accessible from all main screens E5.
- **Security (SEC-02):** The system shall implement an automated session timeout after a period of inactivity to mitigate risks on shared devices ASSUMPTION.

5.4. Prioritization Table (MoSCoW)

ID	Requirement Description	Priority
FR-01	Browse/Filter by materials	Must Have
FR-02	Offline caching of activities	Must Have
FR-03	Session planning with safety notes	Must Have
FR-04	Inclusion prompts for activities	Must Have
FR-07	Data synchronization on signal detection	Must Have

QR-02	Data persistence during signal loss	Must Have
PR-01	Data minimization	Must Have
SEC-01	Secure Sign-Out	Must Have
FR-06	Kit returns tracking	Should Have
FR-08	Manager View/Export	Should Have
SEC-02	Automated session timeout	Should Have
QR-01	Low-spec device performance	Should Have

6. Traceability and Reflection

6.1. Traceability Matrix

Req.ID	Requirement	Priority	Evidence/Assumption	Stakeholder need	Acceptance Indicator
FR-01	Filter by materials	Must	**[E1]**	Facilitator	Returns relevant "Cardboard" results in < 2s while offline.
FR-03	Safety notes	Must	**[E3]**	Volunteer	Prevents session finalization if the safety field is null.
FR-04	Inclusion prompts	Must	**[E6]**	Manager/Pupil	"Inclusion" field appears in every plan template.
FR-05	Aggregate counts	Must	**[E2]**	Coordinator	The system accepts integers for "Category" but rejects text names.
FR-08	Manager View	Should	**[E6]**	Manager	Plan summary is viewable without full facilitator login.

SEC-01	Secure Sign-Out	Must	**[E5]**	Facilitator	Local session tokens are cleared immediately on click.
SEC-02	Auto-timeout	Should	**[ASSUMPTION]**	Coordinator	Session expires after 5 mins of idle time on a shared device.
QR-02	Signal Resilience	Must	**[E5]**	Facilitator	Unsynced draft persists after a hard app-crash

6.2. Critical Reflection

The requirements analysis for STEMMate Namibia highlights the unique challenges of designing for "intermittent connectivity" and "shared hardware." From my observation, I recognize that the synthetic evidence (E1–E6) provides a strong foundation for operational requirements, but significant uncertainties remain regarding "diverse language readiness." I have assumed that plain English and icon-based navigation will suffice, but without linguistic mapping of the specific regions served, there is a risk that the system remains inaccessible to some facilitators. From the perspective of an Ethical Systems Architect, the "shared device" constraint E5 is the most critical security risk. While the report suggests a "secure sign-out," the reality of a busy school environment makes it highly likely that a user will forget to log out. Therefore, I have introduced an automated session timeout (SEC-02) as a necessary mitigation. This adds complexity but aligns with the non-negotiable ethical boundary of protecting data. Another gap is the definition of "low-specification." To move beyond proto-personas, we require technical benchmarks specifically the typical RAM and Android/browser versions used by facilitators in Namibia. This evidence is needed to move QR-01 from a general "Should Have" to a testable technical specification. Finally, the tension between the Equipment Custodian E4 and the Facilitator E1 suggests that the synchronization logic (FR-07) must include a local notification for "Kit Conflict" to prevent volunteers from arriving at a session without the necessary materials. Future research should prioritize field observations of the "kit handover" process to refine these logistical requirements.